(ii) Equation of circle with centre (*h, k*) and radius *r*

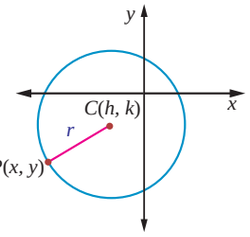


Fig.5.8

Let the centre be $C(h,k)$ and r be the radius and $P(x,y)$ be the moving point.

Then, $CP = r$ and so $CP^2 = r^2$.

That is, $(x-h)^2+(y-k)^2 = r^2$. This is the equation of the circle in Standard form, which is also known as centre-radius form.

Expanding the equation, we get

$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0$.

Taking $2g = -2h, 2f = -2k, c = h^2 + k^2 - r^2$, the equation takes the form

$x^2 + y^2 + 2gx + 2fy + c = 0$, called the general form of a circle.

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ is a second degree equation in x and y possessing the following characteristics:

(i) It is a second degree equation in x and y,

(ii) coefficient of x^2 = coefficient of $y^2 \neq 0$,

(iii) coefficient of $xy = 0$.

Conversely, we prove that an equation possessing these three characteristics, always represents a circle. Let

$$ax^2 + ay^2 + 2g'x + 2f'y + c = 0 \quad \ldots (1)$$

be a second degree equation in x and y having characteristics (i), (ii) and (iii), $a \neq 0$. Dividing (1) by a, gives

$$x^2 + y^2 + \frac{2g'}{a}x + \frac{2f'}{a}y + \frac{c'}{a} = 0. \quad \ldots (2)$$

Taking $\frac{g'}{a} = g, \frac{f'}{a} = f$ and $\frac{c'}{a} = c$, equation (2) becomes $x^2 + y^2 + 2gx + 2fy + c = 0$.

Adding and subtracting g^2 and f^2, we get $x^2 + 2gx + g^2 + y^2 + 2fy + f^2 - g^2 - f^2 + c = 0$

$$\Rightarrow (x+g)^2 + (y+f)^2 = g^2 + f^2 - c$$

$$\Rightarrow (x-(-g))^2 + (y-(-f))^2 = \left(\sqrt{g^2+f^2-c}\right)^2$$

This is in the standard form of a circle with centre C $(-g,-f)$ and radius $r = \sqrt{g^2+f^2-c}$. Hence equation (1) represents a circle with centre $(-g,-f) = \left(\frac{-g'}{a}, \frac{-f'}{a}\right)$ and radius

$$= \sqrt{g^2+f^2-c} = \frac{1}{a}\sqrt{g'^2+f'^2-c'a}.$$

Note

The equation of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ with centre $(-g,-f)$ and radius $\sqrt{g^2+f^2-c}$ represents.

(i) a real circle if $g^2 + f^2 - c > 0$;

(ii) a point circle if $g^2 + f^2 - c = 0$;

(iii) an imaginary circle if $g^2 + f^2 - c < 0$ with no locus.

12th_Maths_EM_Vol1_CH 5_Analytical Geometry.indd 174 16-12-2022 13:27:54